Optimizing sampling and monitoring of species interactions within Biodiversity Observation Networks

Gabriel Dansereau[1,2], Michael D. Catchen[1,2], Ceres Barros[3,4,5] and Timothée Poisot[1,2]

[1] Université de Montréal, Département de sciences biologiques, Montréal QC, Canada

[2] Québec Centre for Biodiversity Science, , Montréal QC, Canada

[3] Canadian Forest Service, Natural Resources Canada, Victoria BC, Canada

[4] Department of Forest Resources Management, University of British Columbia, Vancouver BC, Canada

[5] Département des sciences du bois et de la forêt, Université Laval, Quebec QC, Canada

Correspondence to Gabriel Dansereau — gabriel.dansereau@umontreal.ca

Abstract: Optimal monitoring strategies should be designed to efficiently monitor all essential facets of biodiversity. Yet, species interactions are often overlooked in monitoring designs compared to spatial coverage and species richness, partly due to the inherent difficulty of sampling and monitoring interactions compared to species distributions. Here, we used simulations to test the efficiency of monitoring species interactions and species richness within Biodiversity Observations Networks (BONs). We tested several methods for designing and optimizing BONs to monitor species interactions and examined efficiency when increasing the number of monitored sites. We showed that the required sampling effort is considerably higher for interactions, especially when considering that interaction realization and detection realistically depend on species abundance. Thus, expectations to monitor a relevant proportion of interactions within a BON should be lower than expectations to monitor its species. However, from a single-species perspective, optimizing monitoring designs based on a known species range proved an efficient strategy to retrieve its interactions, outperforming designs based on total species or interaction richness. Only optimizations based on detailed knowledge of where realized interactions occurred allowed bet`ter performance, highlighting the need for better knowledge of where interactions are likely to take place and integration of factors influencing interaction realization, such as species abundances, for further efficiency gains. Therefore, our results offer insights into defining realistic expectations for interaction monitoring. With the right target and

optimization strategy, available information, such as species ranges, can guide observation network designs to monitor species interactions as central biodiversity components.

Keywords: biodiversity monitoring, species interaction networks, biodiversity observation networks, sampling design

Introduction

Biodiversity monitoring should encompass all facets of biodiversity. Article 7 of the Convention on Biological Diversity (United Nations 1992) calls for Parties to "[i]dentify components of biological diversity important for its conservation and sustainable use", and "[m]onitor, through sampling and other techniques, the components of biological diversity". Biodiversity Observation Networks (BONs) are promising initiatives to improve monitoring across scales and biodiversity facets. BONs are coordinated monitoring networks designed to harmonize observation systems, assist reporting on international assessments, and reinforce scientific basis in biodiversity monitoring (Navarro et al. 2017; Gonzalez et al. 2023). BONs can be thematic (Marine BON, Freshwater BON), national (China BON, French BON, Colombia BON) or regional (Asia-Pacific, Europe); some represent networks of monitoring sites for systematic monitoring (China BON) while others represent biodiversity data hubs (French BON) (Navarro et al. 2017). BONs aim to monitor biodiversity change, informing on both current status and trends (Gonzalez et al. 2025). It has been suggested that BONs should eventually be interlinked into the Global Biodiversity Observation System (GBiOS), a globally connected network providing capacity to monitor biodiversity change (Gonzalez et al. 2023), and guide actions needed for targets and goals of the Kunming-Montreal Global Biodiversity framework (CBD 2022). Additional BONs are currently in the development or planification stages, notably in Europe (Kissling et al. 2024) and in Canada (Gonzalez et al. 2025).

Recent work highlighted the need for improved sampling designs in BONs. Current sampling designs over large scales are often too sparse, at too coarse a scale, too infrequent, and without temporal replication (Santana et al. 2025). Recommendations for more effective spatial sampling designs in Europe included i) stratified random sampling (with stratification across environmental, anthropogenic and political gradients or classes), ii) incorporation of existing monitoring sites, iii) filling of spatial gaps, iv) and co-location of monitoring activities (Kissling et al. 2024). Ecological monitoring design relies on site selection algorithms more often discussed regarding their capacity to achieve spatial balance and representative samples, and less frequently with an evaluation to performance to achieve an ecological monitoring goal (Norman and Poisot 2025). For instance, species richness is the most commonly used biodiversity measure in monitoring contexts (Hillebrand et al. 2018), including in discussions over sampling designs, because maximising species richness can be an efficient strategy to design conservation networks while also capturing functional and phylogenetic diversity

(Willig et al. 2023). However, the use of species richness to measure biodiversity change and guide biological conservation has been criticized, mainly since trends in species richness do not properly capture local changes through time as diagnostic indicators should (Hillebrand et al. 2018, 2025), while also neglecting species identities and functional roles and not providing information on persistence of biodiversity and ecosystem services (Aguiar et al. 2024; Fletcher Jr. et al. 2025).

Species interactions are a central component of biodiversity (McCann 2007; Valiente-Banuet et al. 2015; Harvey et al. 2017), can mediate community and ecosystem responses to change (Garzke et al. 2019), and have their own spatial and macro-ecological dynamics (Baiser et al. 2019; Mestre et al. 2022; Windsor et al. 2023; Dansereau et al. 2024). We now have sufficient conceptual tools to support the added value of species interactions in designing efficient management actions (Dansereau et al. 2025; Moracho et al. 2025). However, species interactions are still considered to undergo a biodiversity shortfall, due to a lack of data at global scales (Hortal et al. 2015), as well as the uneven data availability across different interaction types, geographical locations and environments (Poisot et al. 2021). Species interactions represent a critical component of ecosystem functioning and health, yet may be lost (or change) faster than species (Valiente-Banuet et al. 2015). Monitoring interactions is more informative regarding ecosystem functions than monitoring species richness (Jordano 2016). Network responses may be different from species richness (e.g. Doré et al. 2021), emphasizing the need for ecological monitoring to consider interactions, not just species richness. Species interactions can also improve our ability to predict responses to climate change (Abrego et al. 2021). For both these reasons, sampling designs should incorporate considerations for species interactions.

Prior investigations of interaction sampling did not address how efficiently we should expect to monitor interactions within the BON framework, nor did studies on the effectiveness of BONs pay particular attention to the sampling of species interactions. Instead, much focus has been given on how to assess sampling completeness (e.g. Chacoff et al. 2012) and how sampling effort affects network properties (e.g., McLeod et al. 2021). Recent empirical (Caron et al. 2024) and theoretical (Poisot 2023) results have established that knowing the network structure does not always allow to know the list of interactions (and vice versa), therefore justifying the need to sample interactions as their own biological entities. A central challenge with the sampling of interactions is that the sampling effort required to document interactions is much higher than for species (e.g. 500% vs 64% in Chacoff et al. 2012). Also, evaluating sampling completeness for interactions is more complex than for species. In addition to the analysis of sampling robustness, it requires assessing the proportion of forbidden links (i.e. links that cannot occur) due to life history or morphological restrictions, which in turn requires knowledge of species natural history and traits (Jordano 2016). While some quantitative network properties such as connectance are less affected by sampling completeness, network composition and binary measurements derived from it (e.g. number of links, Vizentin-Bugoni et al. 2016; Poisot et al. 2012) will vary widely if interactions are missed. Small networks are also more sensitive to undersampling than larger ones (Henriksen et al. 2019). Quantitative frameworks exist to evaluate network taxonomic, functional and phylogenetic diversity from incomplete samples (Chiu et al. 2023).

A much needed element is to consider sampling network and interactions from an explicitly spatial perspective with variation in local sampling structure —that is, quantifying sampling effort from specific sites where interactions are sampled. For instance, Poisot et al. (2012) measured sampling effort based on resampling a dataset to document its effect on network composition and properties. Other examples focus on scale, notably showing that many network structure metrics are robust across spatial scales when accounting for expected increases of richness and decreases in connectance (Wood et al. 2015). In contrast, McLeod et al. (2021) used data from specific lakes compare sampling strategies, showing that sampling larger to smaller lakes better captures regional metaweb properties. We now have frameworks and models to describe and investigate the variation and detection of interaction networks in space and time (Poisot et al. 2015; Cirtwill et al. 2019; Catchen et al. 2023), and therefore the next step is to consider how to optimize monitoring for species interactions. Simulations have already been used in this context to: generate communities and evaluate the potential of metrics to track biodiversity change (Santini et al. 2017); simulate networks and sampling to evaluate sampling effects on estimated network properties compared to a known reference (de Aguiar et al. 2019, although not in a spatial context); determine optimal sampling frequency for monitoring (Daugaard et al. 2024); examine the dependence of monitoring efficiency and trends detection on the number of monitoring sites, species abundance, detection probability and strength of decline, under a range of scenarios and monitoring schemes (Ficetola et al. 2018).

In this manuscript, we address three key questions. First, how should we implement monitoring programs across space to efficiently capture network composition and changes, as essential components of biodiversity? Second, can we monitor species interactions as efficiently as species ranges? Finally, what effectiveness can we realistically expect from sampling designs optimized to monitor species interactions?

We use spatially explicit community simulations to test the efficiency of monitoring species interactions within BONs, and explore various algorithmic and design options for network sampling optimization. We aim to define realistic expectations for monitoring species interactions, in comparison to species richness and species ranges. To do so, we compare design and optimization options, examining how to assess and compare their efficiencies. We conclude with recommendations on how this should influence the design of BONs to improve the monitoring of species interactions as an important biodiversity component.

Methods

We used simulations to generate species interaction networks, species ranges, and biodiversity observations networks (BONs) with the goal of evaluating monitoring potential for interaction networks within BONs. We first detail the steps for the general model we used (Figure 1). We then go over three simulation studies we performed to investigate specific elements related to the monitoring of species interactions.

General model

We followed the approach and process-based, spatio-temporal model introduced by (Catchen et al. 2023; building on Cirtwill et al. 2019) to simulate interaction realization and detection while accounting for spatial variation in species occurrences, implemented in SpeciesInteractionSamplers.jl. This model broadly captures the different processes involved into turning a *potential* interaction (i.e. a biologically feasible interaction) into a *realized* one (i.e. an interaction that took place at a specified location), as outlined in Morales-Castilla et al. (2015). To this generative model, we added the design of Biodiversity Observation Networks and the evaluation of sampling efficiency between and across simulations for species interactions (Figure 1A).

Step 1: Generating interaction networks with spatial variation

First, we simulate networks with variation in species and interaction composition across a simulated landscape (Figure 1A, Step 1). To do so, we generate autocorrelated species ranges representing the distribution of species across the landscape using neutral landscape models [NLMs; Gardner et al. (1987); Etherington et al. (2015)]. NLMs are used to generate landscapes replicating realistic autocorrelated distributions without underlying biotic or abiotic factors,

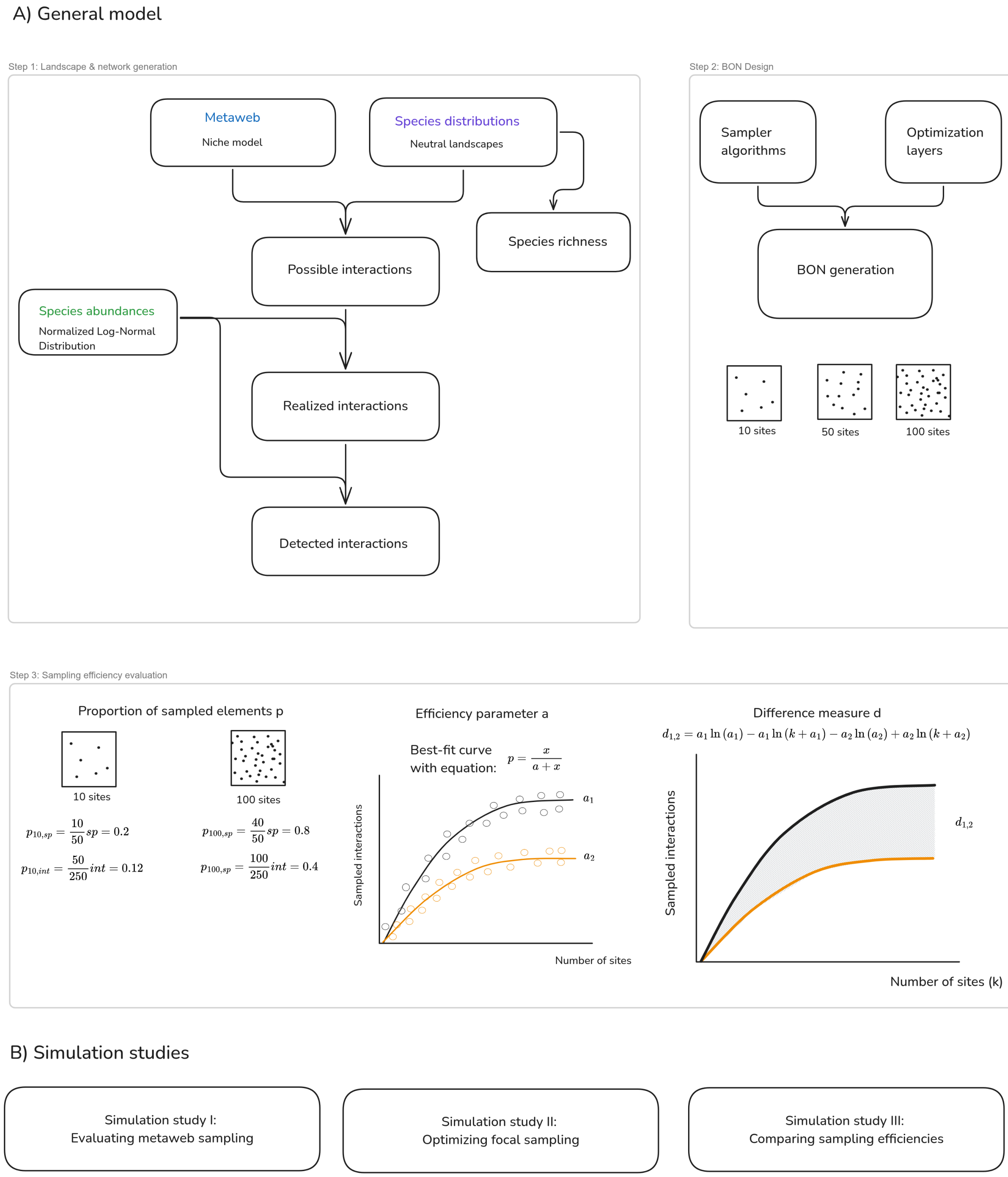

Figure 1: Conceptual diagram of the workflow and simulations. The general model (A) was used for the simulations, starting from generating landscapes with spatial variation in species and interaction composition, designing biodiversity observation networks (BONs) and measuring sampling efficiency within and across simulations. The simulation studies (B) use the general model to investigate specific elements related to sampling interactions with BONs.

and therefore represent useful baselines to explore ecological processes in a spatial context (Hesselbarth et al. 2024). NLMs have been used to generate realistic landscapes in contexts relating to biodiversity distribution and preservation, for instance to compare avian diversity between conservation approaches in coffee-growing landscapes (Valente et al. 2022) or to assess the potential of a suggested umbrella species in an ecotone (Duchardt et al. 2023). NLMs have also been used with species interactions, for example to evaluate the effect of landscape structure in cross-species disease transmission between interacting species (Forero-Muñoz et al. 2025). Here, we use NLMs to create communities of known composition and species richness at every location in our simulated landscape.

Next, using the spatio-temporal model presented by Catchen et al. (2023), we define the **feasible interactions** between the species by generating a metaweb using the niche model (Williams and Martinez 2000). The niche model defines interactions by assigning species a niche position and feeding range, similarly to body-size relationships between predator and prey, and generates networks with structural properties similar to empirical food webs (Williams and Martinez 2000; Gravel et al. 2013; Delmas et al. 2019). It is widely used as a generative model, including in the context of investigating interaction sampling (Poisot et al. 2012; Wood et al. 2015).

Combining the species ranges and the metaweb, we verify where species with feasible interactions co-occur, to build a list of **possible interactions** (Catchen et al. 2023) at every location in our landscape. Next, we define **realized interactions** by conditioning possible interactions to species abundances. For example, rare and low-abundance species are less likely to interact in a location because they are less likely to encounter one another when compared to abundant species, leading to neutrally forbidden links (Canard et al. 2012; Catchen et al. 2023). The neutral model introduced by Catchen et al. (2023) first generates a distribution of species relative abundances from a log-normal distribution, then models interaction realization as a stochastic Poisson Process, drawing realized interactions from a Poisson distribution parametrized on the relatives abundances of interacting species and a fixed realization energy parameter (see Catchen et al. 2023 for full details).

Finally, we define the set of **detected interactions** by also conditioning the detection of realized interactions on species abundances. Even if an interaction is realized in a location, sampling must occur at the moment of interaction for its detection; this is again less likely for low probability events, like interactions between rare occurring species; see Catchen et al. (2023) for implementation). The result of this model is a landscape with spatial variation in species and interaction composition and detection, where species composition, realized interactions, and detected interactions are based on the process model and known at all locations, allowing to compare the efficiency of sampling strategies.

Step 2: Designing Biodiversity Observation Networks

We design Biodiversity Observation Networks (BONs) across the simulated landscape and measure their efficiency at monitoring the generated interaction networks. Norman and Poisot (2025) presented an overview of the design and site selection process accounting for ecological monitoring goals. The design process uses a site selection algorithm to designate candidate sites for the monitoring network. Key features of the algorithms include the ability to balance sites across space for equal representation (see Benedetti et al. 2017; Kermorvant et al. 2019 for reviews), to stratify across regions or populations, to include auxiliary site information (environmental or biological) to weight inclusion probability, and to produce master samples including current monitoring site or from which to select a subset of sites for a specific monitoring project (van Dam-Bates et al. 2018; Robertson et al. 2024).

Using our model simulations, we can use different algorithms to distribute monitoring sites across the landscape, while optimizing their location given different information such as species ranges, species richness or presence of realized interactions, and compare algorithm sampling efficiency. For example, we can design BONs using the Balanced Acceptance Sampling algorithm (Robertson et al. 2013) when the aim is to balance sites across our simulated landscapes, given its computational efficiency and its ability to weight site inclusion probabilities using a layer of weights. On the other hand, active learning methods like the Uncertainty Sampling algorithm (Lewis and Gale 1994) can be used to optimize sampling given information on element uncertainty (Settles 2009). Instead of focusing on uncertainty, we can apply the algorithm to target the sites that provide the highest information gain for the BON, given a layer of 'information' values to maximize. Examples of active learning relating to biodiversity monitoring and sampling site selection include improving species range estimation from a smaller number of actively selected sites (Lange et al. 2023), minimizing field data collection costs based on distances between samples (Malek et al. 2019), and identifying uncertain labels for classification to improve bioacoustic monitoring performance across stratified samples (McEwen et al. 2025).

Step 3: Evaluating sampling efficiency with comparable measures

Our simulations require evaluating sampling efficiency at three levels (highlighted in Figure 1, Step 3): at a landscape scale for a given BON and number of sites (a snapshot description measure), as an efficiency curve describing the efficiency with an increasing number of sites (a snapshot ensemble summary measure), and as a difference score to compare optimization options (an ensemble comparison measure).

On the first level, we use the **proportion of sampled elements** (species or interactions, noted p) to evaluate how efficient a given BON is on a landscape scale. In simulated landscapes, species and interaction composition are known everywhere and thus we can determine the proportion of species and interactions that can be sampled for each generated BON (e.g. a BON made of 50 spatially-balanced sites). This proportion of sampled elements compared to the true richness (of species or interactions) can be seen as a 'snapshot' view of a given BON's sampling efficiency for that simulated landscape.

On the second level, we define an **efficiency parameter** (a) describing how sampling efficiency changes as the number of sites increases in a given BON. Ideally, we need one summary measure to summarize BON-efficiency for a given landscape and optimization strategies across the ensemble of snapshots. Moreover, this measure needs to be comparable across samplers, optimization layers and species degrees (number of interactions), given how the generated metawebs and landscapes might vary between independent simulations. Since the exact proportion of monitored interactions is known in our simulations, we used the following single-parameter equation which generates similar efficiency curves:

$$p = \frac{x}{a + x} \tag{1}$$

where p is the proportion of monitored interactions, x is the number of sites in the BON and a is the efficiency parameter we use for comparison. Here, a is the single parameter describing the shape of the curve, and can be compared across samplers, optimization layers or focal species degree. The lower this parameter, the faster we reach a high proportion of monitored interactions. We can identify the value of a yielding the curve with the best fit to the observed data, as described in our Simulation study III.

On the third level, we define a **difference measure** (d) to compare optimization algorithms and layers for the same simulated landscape. We have efficiency curve parameters a, which we can compare between options on a greater or lower basis, but the order of magnitude of the difference is not necessarily comparable across sets of simulations, hence the need for a more formal measure. To compare the efficiency curves within a given simulation (i.e., between samplers and optimization layers for a given metaweb and set of species ranges), we used the definite integral of Equation 1 between 0 and k, the maximum number of sampling sites in the landscape (10,000):

$$\int_0^k \frac{x}{a + x}\,\mathrm{d}x = a\ln(a) - a\ln(k + a) + k \tag{2}$$

and computed the difference in efficiency between any two curves as the difference of integrals:

$$d_{1,2} = a_1 \ln(a_1) - a_1 \ln(k + a_1) - a_2 \ln(a_2) + a_2 \ln(k + a_2) \tag{3}$$

with positive difference values indicating that the first curve is more efficient than the second. This difference measure allows identifying which samplers or optimization layer are more efficient – i.e. allow reaching a high proportion of monitored interactions more quickly – while also quantifying the magnitude of this difference in a way that is comparable between sampling strategies. We can then compare all samplers and optimization layers individually within each independent simulation, given that the monitoring context is different in each one (different metawebs and species ranges).

Simulation studies

We performed three simulation studies to investigate the efficiency of monitoring species interactions within BONs. Across our simulations, we used landscapes of 100 x 100 pixels, 75 species, and a connectance of 0.2 to generate metawebs using the niche model. The connectance and number of species are within the range of values of well-characterized empirical food webs (Williams and Martinez 2000; Dunne et al. 2002, 2004; Smith-Ramesh et al. 2017) commonly used for reference (Williams and Martinez 2008; Baiser et al. 2010; Wood et al. 2015). These parameters are also within the range of previous simulation studies on spatial ecological network dynamics, *e.g.* Thompson and Gonzalez (2017). Exploratory simulations showed the number of sites in the BON to be the main element affecting the proportion of sampled species or interactions compared to other model parameters such as metaweb connectance and landscape autocorrelation. The activation energy parameter expectedly influenced results as covered in Catchen et al. (2023) when presenting the model.

Simulation study I: Efficacy of spatially balanced sampling for metawebs vs. species richness

Our first simulation study aimed to investigate the efficiency of sampling networks across space to document a metaweb of interactions, when compared to documenting species richness. We generated spatially balanced BONs across our simulated landscape that defined sites to sample species and interactions. We varied the number of sites between 1 and 100. We then evaluated the sampling efficiency of each BON in terms of sampled species richness and metaweb interactions. For species richness, we assumed that all species present in a sampled location were sampled and compared the number of species sampled across all sampled sites against the total species richness across the landscape (always 75 species). For interactions, we aggregated each interaction type separately (possible, realized and detected) for a given BON and evaluated the proportion of the metaweb interactions documented through sampling. We ran simulations with 100 fully-independent replicates (different metaweb, species ranges, abundance distribution, realized interactions) for each number of sites in a BON, yielding 10,000 simulations in total.

Simulation study II: Optimizing focal sampling to monitor a given species' interactions

Interaction network sampling and monitoring may be targeted at a focal species because of its endangered status (e.g. Rioux et al. 2022), because it is at the centre of conservation or restoration programmes (e.g. BearConnect), or even due to human-nature conflicts (e.g. Baranowska et al. 2025). Hence, we also investigated sampling efficiency from a focal species perspective, by simulating a virtual observer that would aim to sample all the interactions of a focal species (here known from the metaweb), but might have different information on how to optimize the sampling design. Observation network design and efficiency needs to be evaluated beyond spatial balance, and also regarding their efficiency at meeting biodiversity monitoring objectives (Norman and Poisot 2025). Thus, the idea of our focal sampling exploration is to represent a simplified case with a clear objective (retrieving a species' interactions) and where we do have prior knowledge about the system, either through empirical data or model predictions, which can be used to optimize sampling.

Optimizing focal sampling relies on two elements: the sampler algorithm used and the information provided for optimization. Species range represents an attainable level of information an interested observer could have, for instance based on expert knowledge or species distribution models. Therefore, we assumed the observer knew the species range across the landscape (presence or absence). We tested three sampling algorithms for comparison: simple random sampling (to represent uninformed sampling), weighted balanced acceptance sampling (aiming to balance sites across space while weighting favorably for species ranges) and uncertainty sampling (which targets locations with higher 'information' values in the layer provided). Next, we tested four information layers for optimization: one with the focal species range (our attainable information knowledge for the observer), one with the probabilistic range (for a less informed case, similar to the output of a species distribution model), one with the location of realized interactions for the focal species (to represent a best-case scenario) and one with species richness across the landscape (to represent a holistic level of information about the system).

Using the approach described in Step 1 of the the General Model, we generated one set of metaweb, species ranges, and locations for realized interactions (here in Simulation study II we focused on a single set to showcase a sampling design example across a given landscape, and we counter-verified it with separate sets which yielded conceptually similar results; see Simulation study III for a comparison of several landscapes). We selected the species with the highest degree (number of interactions) in the metaweb as the focal species for the simulation [as it proved as an efficient strategy for sampling and conservation; Pires et al. (2017); de Aguiar et al. (2019)]. We then generated BONs of 1 to 500 sites (at interval increments of 5 sites) using the three possible algorithms (simple random, weighted balanced acceptance, uncertainty sampling) and four optimization layers (focal species range, probabilistic range, realized interactions, species richness), with 100 replicates for every number of sites. For each BON generated, we extracted the number of unique realized interactions sampled across all sites (the sampled degree for the focal species).

Simulation study III: Comparing sampling efficiency across different landscapes and metawebs

We verified whether our results were constant across simulations that differed in the generated metawebs and species ranges. We ran 100 independent simulations, each with 50 replicates for every number of sites (once again from 1 to 500 sites at interval increments of 5 sites). Doing so raised the issue of describing and comparing the efficiency of the simulated monitoring across samplers, optimization layers or focal species degree, as generated metawebs and landscapes varied between independent simulations. The comparison measures described earlier in Equation 1 and Equation 3 were used to provide comparable measures of efficiency. Namely, the efficiency parameter a (Equation 1) was used to describe the efficiency of all independent simulations, and the difference measure d (Equation 3) was used to compare the efficiencies of different optimization options within a specific landscape and simulation.

For each independent simulation, sampler and optimization layer option, we performed a grid search across possible values of a (from Equation 1) to identify the curve yielding the best fit with the simulation's result. We searched for parameter values on an exponential scale as it allowed for more gradual control over the curve shape, evaluating 10,000 evenly spaced values between e^{-5} and e^{15} (limits chosen to go beyond the fitted values), and selected the value of a yielding the curve minimizing the sum of squared errors to the simulation results.

We compared all samplers and optimization layers individually within each independent simulation, given that the monitoring context is different in each one (different metawebs and species ranges), using our difference measure d (Equation 3). The measure allows to identify which samplers or optimization layer is more efficient, thus quickly reaching a high proportion of monitored interactions (positive values indicate that the first curve is more efficient than the second), while also quantifying the magnitude of this difference in a comparable way between possible comparison pairs.

Results

Simulation study I: Lower sampling expectations for species interactions than species richness

Our results for metaweb sampling highlight a marked difference between the proportions of sampled species and interactions (Figure 2). Sampled species saturated very quickly, with all species sampled with less than 25 sites in a BON across the landscape. Possible interactions, which only depend on co-occurrence of species with a feasible interaction in the metaweb, also saturated quickly reaching over 90% of interactions in the metaweb with 25 sites and increased asymptotically afterwards. In contrast, realized and detected interactions, which account for the impact of species abundances on interaction detectability, reached much lower proportions, both remaining under 25% of the all metaweb interactions at 100 sampling sites. While the proportion of sampled realized and detected interactions would keep increasing with a higher number of sites, the speed at which they accumulate is notably lower than for species richness or possible interactions, highlighting the need for a much higher sampling effort.

Simulation study II: Differences between samplers and optimization layers

Our focal sampling experiment showed the Uncertainty Sampling algorithm to be most efficient at optimizing BON-design based on a known species range and while aiming to maximize the sampling of interactions of a single focal species (Figure 3A). Simple Random sampling, as a proxy for uninformed sampling, performed the worst among the algorithms tested. Weighted Balanced Acceptance performed notably worse than Uncertainty Sampling at maximizing the efficiency for the focal species, although it maintained a better spatial balance across the landscape.

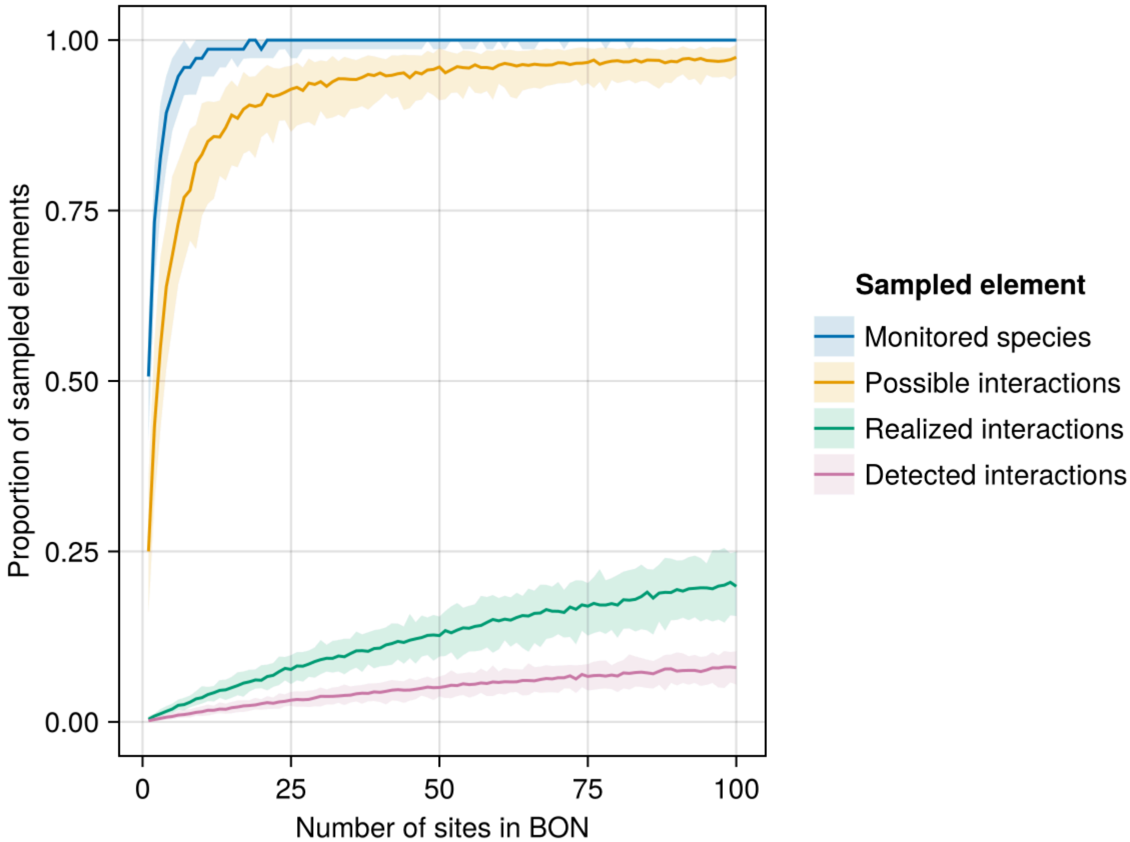

Figure 2: Proportion of sampled species and interactions for a given number of sampling sites in a Biodiversity Observation Network (BON) across the simulated landscape (Simulation Study I). Proportions were calculated with respect to the total species richness and the total number of interactions in the metaweb. Lines indicate the median proportion for a given number of sites, while the shaded bands delimit the 5th and 95th percentiles across replicates (100 per number of sites; 10,000 in total).

Optimizing on the known species range, our proxy for an attainable level of information an interested user might have, was more efficient at sampling the focal species' interactions than optimizing based on species richness, effectively sampling more than half of the species' interactions (Figure 3B). In contrast, optimization based on the location of realized interactions (our best-case scenario requiring a higher level information) expectedly performed better, retrieving most of the species' interactions.

Simulation study III: Comparison of efficiency parameters across simulations

The distribution of efficiency parameters across independent simulations confirmed the interpretations based on Figure 3, but also highlighted a wide range of outcomes in efficiency between simulations. Among samplers, the median efficiency parameter indicated a better performance (lower parameter value) for Uncertainty Sampling, followed by Weighted Balanced Acceptance and Simple Random (Figure 4A). The actual efficiency values spanned a similar range across the possible sampler options with all simulations combined. Among

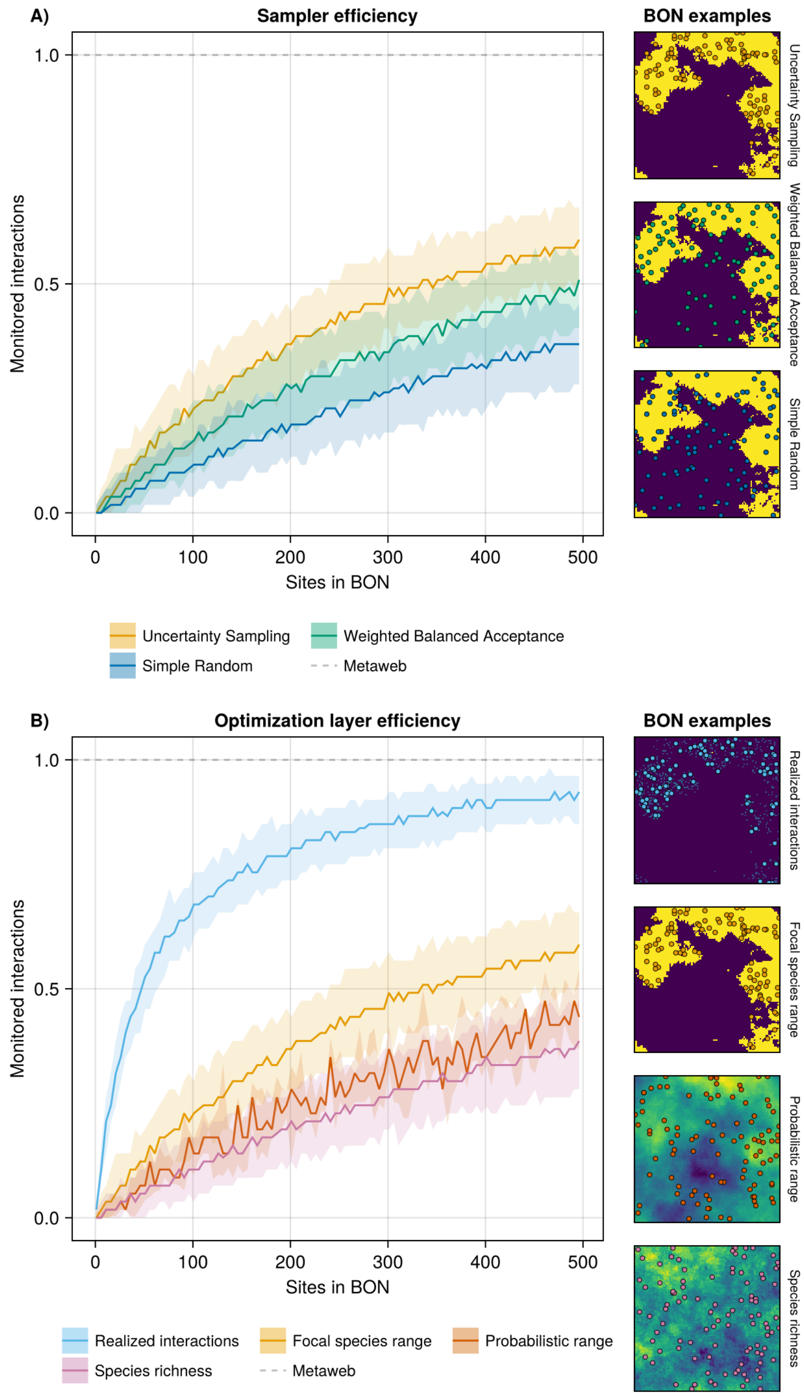

Figure 3: Efficiency of sampler algorithms (A) and optimization layers (B) at sampling realized interactions for a given focal species within Biodiversity Observation Networks (BONs). Panel A compares three samplers with a fixed optimization layer (the species range), while Panel B compares four optimization layers with a fixed sampler (Uncertainty Sampling). Lines indicate the median proportion for a given number of sites, while the shaded bands delimit the 5th and 95th percentile across replicates. The panels on the right illustrate examples of BONs with 50 sites (coloured points) generated by each sampler over the same landscape (species presences are shown as pixels in yellow).

optimization layers used with Uncertainty Sampling, optimizing on the focal species presence/absence range performed better than on the probabilistic range, followed by species

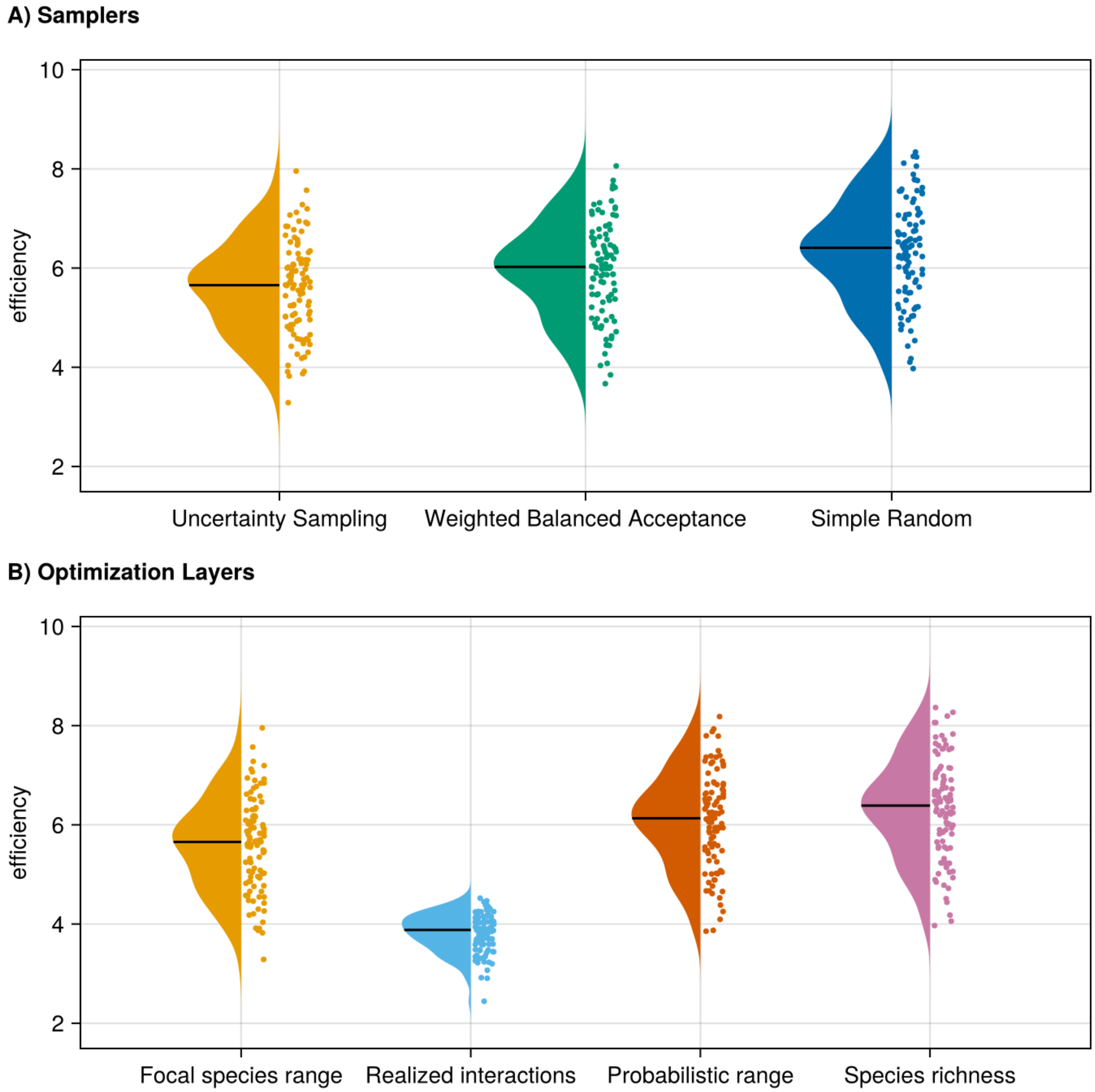

Figure 4: Distribution and density of efficiency parameters across 100 independent simulations for the three sampler options tested on the focal species range (A) and for the four optimization layer options using Uncertainty Sampling as the sampler (B). 'Uncertainty Sampling' on panel A used the focal species range layer, hence it shows the same data as 'Focal species range' on panel B. The efficiency parameter is the natural log of parameter a in Equation 1, describing the best-fit sampling efficiency curve for a given simulation.

richness, although all three options spanned similar ranges with all simulations combined (Figure 4B). In contrast, optimizing based on the realized interactions distinctly and consistently performed better across all simulations.

The fact that multiple options span similar ranges of efficiency values across independent simulations highlights the need to compare the efficiency within a given simulation. Comparing optimization options by simulation clearly highlighted a consistently more efficient approach across landscape and metaweb contexts (Figure 5; see also Supp. Mat.), consistent

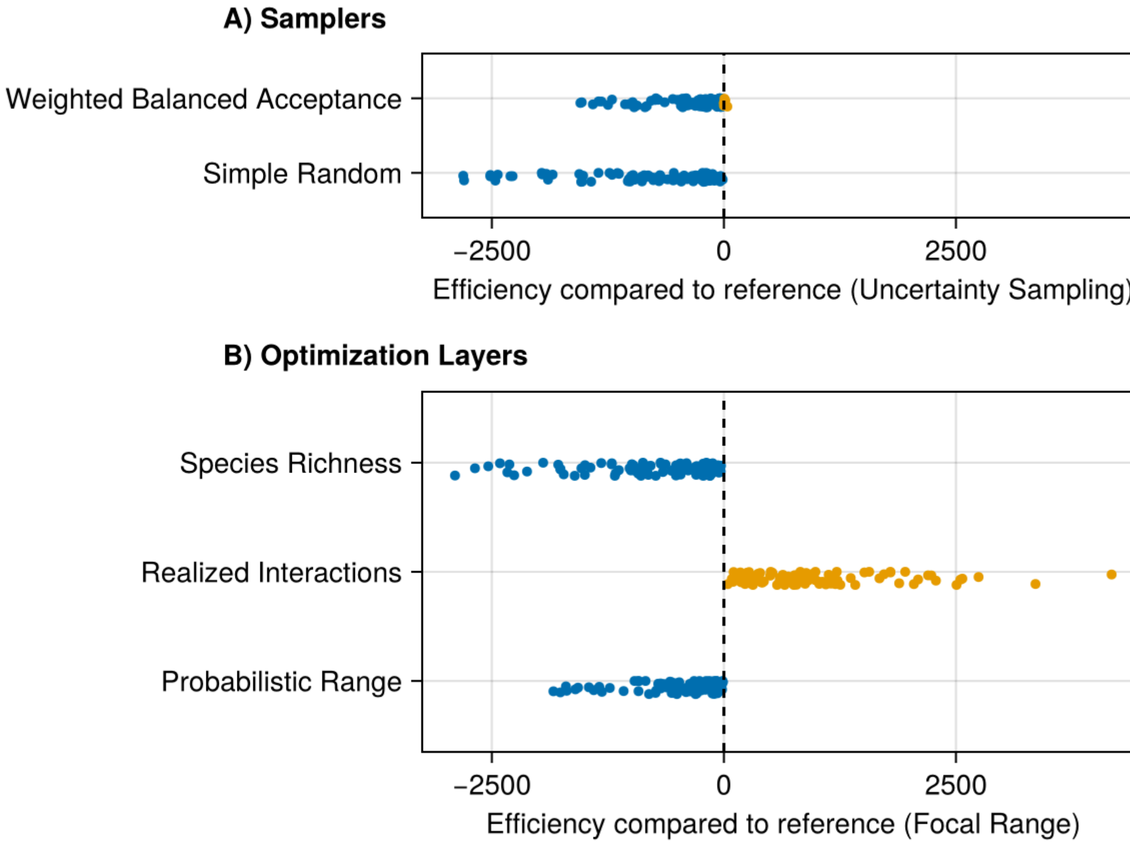

Figure 5: Within-simulation comparison of efficiency between samplers and optimization layers. The comparison value is based on the difference of the definite integrals of the efficiency samplers curves described in Equation 3. Negative values (in blue) indicate that the compared option led to a less efficient sampling than its reference, while positive values (in orange) indicate it was more efficient. References were chosen for narrative purposes and to simplify the number of comparisons displayed. All one-by-one comparisons are available in Supp. Mat.

with the showcased single-simulation example (Figure 3). Uncertainty Sampling was more efficient than Weighted Balanced Acceptance in 98 of 100 simulations, and both were more efficient than Simple Random in all simulations (Figure 5). Optimizing on the focal species range was always more efficient than on the probabilistic range or on species richness (with probabilistic range more efficient than species richness in all but one simulation), but always less efficient than optimizing on realized interactions.

While the simulations clearly highlight which sampler or optimization layer is most efficient across all simulations, efficiency parameters and differences vary over a wide range of values. We verified if this could be tied to the occupancy of the focal species in simulated landscape, measured as the number of cells where the species is present. Optimizing using Uncertainty Sampling on the focal species range and on realized interactions is especially efficient at low occupancy, as it adequately targets the fewer cells where the focal species is present. As species occupancy increases, the difference between optimizing approaches decreases. When the focal species has high occupancy across the landscape, Weighted Balanced Acceptance and Simple Random algorithms perform similarly to Uncertainty Sampling, as balancing sites across space or choosing at random becomes equivalent to targeting the species range. Optimizing based on species richness also became more efficient as occupancy increased. A possible explanation is that at high occupancy, species range is not a restricting element and thus targeting sites with high richness directly leads to targeting sites with potentially more interaction partners. On the contrary, at low occupancy, the species range is an important restriction, and optimizing on species richness might target sites where the focal species is absent. Optimizing on the probabilistic species range resulted in similar outcomes; at high occupancy (not to be confused with probability of occurrence) the sampler can more effectively filter out sites where the species is absent, since it is very likely the species will be present at the sites that have high occurrence probability. At low occupancy, the focal species will be present in fewer locations with high probabilities and might be absent from locations with intermediate probabilities, making it harder for the sampler to target sites where the species is present.

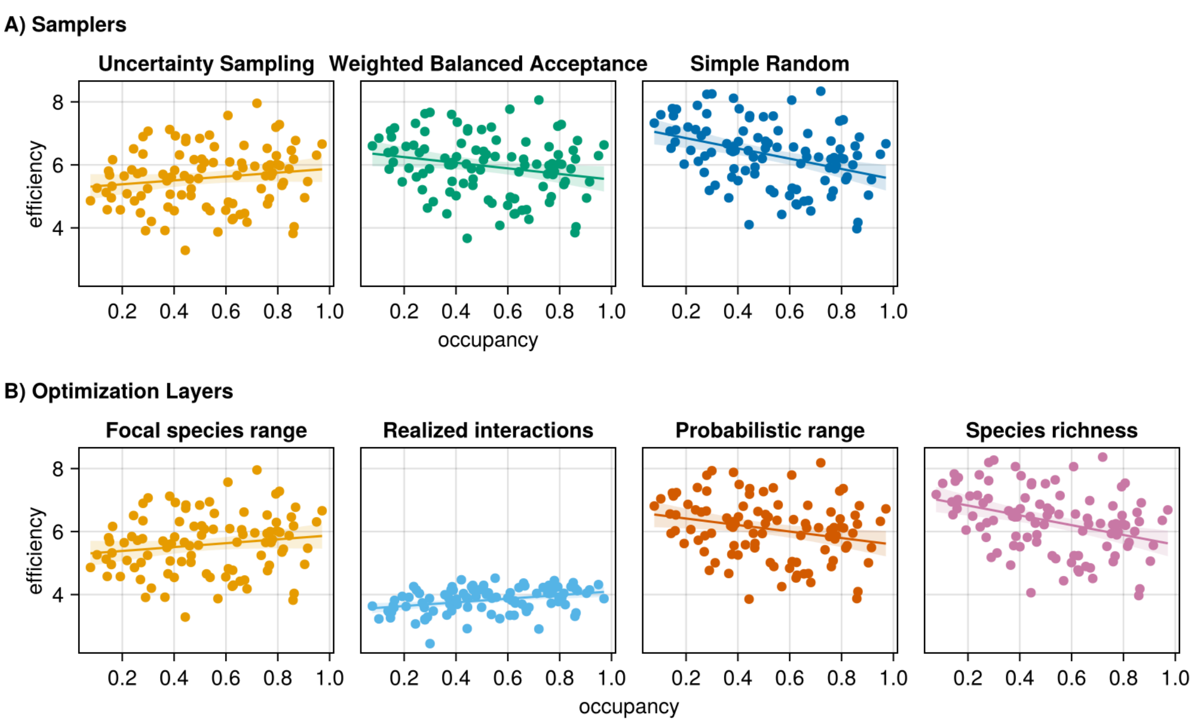

Figure 6: Relationship between the occupancy of the focal species and sampling efficiency for all optimization options. These results come from 100 independent simulations, and therefore 100 different sets of metawebs, species ranges, and occupancy of the focal species. The efficiency parameter is the natural log of parameter a in Equation 1, describing the best-fit sampling efficiency curve for a given simulation.

Discussion

What should be realistic expectations for sampling and monitoring of species interactions?

Our simulations highlight that expectations should be much lower than for species richness, especially when considering that interaction realization and detection both depend on species abundance. In other words, we can expect the monitoring of species interactions within a BON to require a (much) higher monitoring effort across space than the one required for monitoring species and community composition. This is similar to observations regarding sampling effort required to document pollinator networks (Chacoff et al. 2012), yet here framed in the context of BON design and monitoring across space.

We can improve coverage with the right optimization method, but efficiently doing so requires a clear and attainable sampling or monitoring objective. For instance, adopting a single-species perspective [*e.g.* focusing on interactions for a keystone or high degree species; Pires et al. (2017); de Aguiar et al. (2019)] can yield relevant results while relying on attainable levels of information. Optimizing on a known species range proved an efficient strategy to retrieve its interactions that was better than optimizing based on more holistic, community-level information such as species richness (identified in contrast by Willig et al. 2023 as an efficient target to maximize to conserve multiple biodiversity facets). Optimizing based on the location of the realized interactions was, as expected, even more efficient. Yet, this represents a best case scenario that is likely beyond reach in an applied context: it assumes that there is already pre-existing knowledge on where species interactions can be observed, which is unlikely to exist for most interactions, many species and in many locations across the globe, regardless of their relevance for monitoring. Nonetheless, it highlights that the most important efficiency gains will come from better knowledge of where interactions are likely to take place, which suggests that monitoring of interactions will become more efficient over time, especially if local knowledge of the system and species under focus is incorporated. Additionally, better incorporating considerations for species abundance, the factor influencing interaction realization and detection in our simulations, will be crucial and potentially more impactful than improving knowledge or reducing uncertainty over species ranges.

When does the choice of sampler or optimization layer matter the most?

Here we concur with Norman and Poisot (2025) that ecologically-relevant metrics are required to evaluate the performance of monitoring designs, beyond spatial balance. However, while they found that many sampling algorithms performed similarly for biodiversity monitoring, our results highlighted better options when the monitoring target is highly localized. We observed the most important differences for species with low occupancy, where Uncertainty Sampling is relevant to target only the focal species range. Differences were less important at higher occupancy and for species present across the whole landscape, although Uncertainty Sampling still performed better. This result should be expected, as a targeted sampler should be relatively more efficient when a species range is highly localized compared to one balancing sites across the the whole landscape. Nonetheless, it highlights how some highly-localized monitoring targets (e.g., species interactions) might benefit from targeted monitoring designs. In such cases, efficiency gains will depend on improvements in our capacity to predict the location of those phenomena and on our ability to continuously evaluate and update predictions based on the information acquired.

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